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Agroforestry adaptation in individual households in northwestern Vietnam: who needs what and implications for practice and policy

Hoa Do¹*, Eike Luedeling¹, Cory Whitney¹

¹ Department of Horticultural Sciences, Institute of Crop Science and Resource Conservation (INRES), Bonn, Germany

*Corresponding author (s78tdo@uni-bonn.de): Hoa Do is a PhD student in the Horticultural Department at the University of Bonn. She is interested in methodologies for understanding the complexity of agricultural systems and emphasizes an interdisciplinary approach, particularly in farming systems and modeling for agricultural decision support. Her research and professional experiences are centered around agroforestry and community development based on agriculture. For her doctoral research, Hoa is investigating the adoption and adaptation of agroforestry practices in the mountainous regions of northwestern Vietnam.

Abstracts: Agroforestry is promoted in northwestern Vietnam to enhance environmental sustainability and support livelihoods, especially in resource-constrained settings. Its adoption is hindered by farm-level and structural barriers, necessitating interventions tailored to diverse needs of farmers. Employing a participatory approach, we developed a farm typology to understand the heterogeneity among agroforestry farmers by analyzing data from 101 households through archetypal analysis. This analysis identified three distinct agroforestry archetypes, each reflecting unique resource constraints. Archetype 1 (A1) and archetype 2 (A2) operate small-sized farms with limited agricultural labor and diversified livelihood activities. A1 farmers manage concentrated land and maximize land and labor use efficiency by integrating diverse agroforestry systems. In contrast, A2 farmers deal with high land fragmentation and small-sized plots, which restricts the diversification of agroforestry systems. Archetype 3 operates the largest farms, focusing their livelihood on cash crops with the lowest adoption level and moderate species diversity in their agroforestry systems. We used the archetype profiles to discuss with farmers potential interventions to improve their engagement in sustainable agroforestry practices. Strategic interventions include improving market access, providing financial support for inputs and tree seedlings and establishing irrigation systems. Practical interventions involve diversifying agroforestry systems and enhancing technical knowledge. We discuss potential socio-technical solutions and challenges and opportunities for their implementation. Our study emphasizes the importance of tailored interventions to meet diverse farmer needs and contributes to the co-creation and design of farming systems. It facilitates learning from varied farmer behaviors and strategies, ensuring the practicality and local relevance of proposed solutions.

Keywords: Agroforestry adoption, archetypal analysis, sustainable farming practices, participatory approach.